RECONIQ

Verified bank reconciliation

Code does the arithmetic. The model picks. Checks verify.

49% → 96% group F1

10,000 bank lines · 9 models

1 · THE PROBLEM

Reconciliation is exact arithmetic plus judgment

1 deposit → 4 invoices

One to many

3 partials → 1 invoice

Many to one

Payout = sale − fee

Fee netting

3 entries, same amount

Decoys: only the text decides

1% of many-to-one groups found by GPT-4.1 given a whole month. A plausible match that doesn't add up is an accounting error.

Small businesses are the biggest AI productivity gap

43.5%

of US GDP comes from small businesses

<20%

of firms with 4 or fewer employees use AI, and use isn't growing below 20 employees

Largest share of output, slowest adoption. Monthly reconciliation is manual, repetitive and the same in every business: a first workflow to automate.

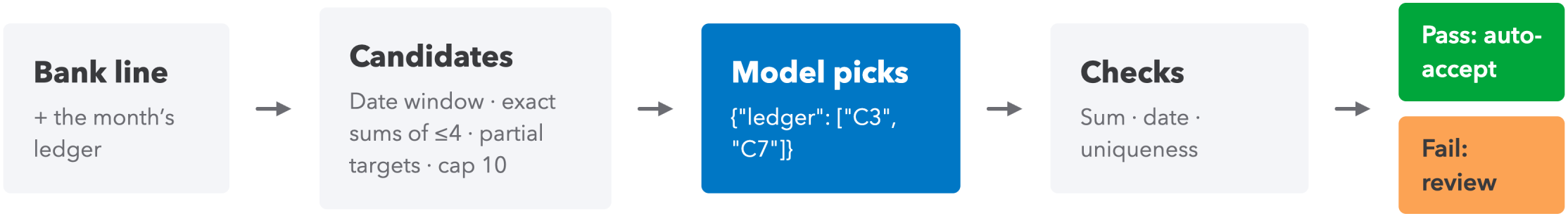
Sources: SBA Office of Advocacy, Small Business GDP 1998–2014 (2014 share) · US Census Bureau, Business Trends and Outlook Survey, Dec 2025–May 2026

Each layer unlocked part of the evaluation

Technology	Role	What it made possible
Ledger-first generator	Exact gold, 5 match types	Exact group F1 on 10,000 lines, no hand labels
Subset-sum + checks	Arithmetic in code	Auto-accepts at 99.95% precision
Fireworks serverless	GPT-OSS-120B, Nemotron	Full 10,000-line run for $1.81
GenOS	GPT-4.1, GPT-5.1	Frontier baselines on the same data
Ollama + MLX	phi4, Llama 3.1, Qwen3	Free local runs and LoRA on a laptop

Impossible without them: exact gold at 10,000 lines can't come from hand labelling, and per-line calls at GPT-4.1 prices cost 8× more.

The harness: code proves, the model picks



The model never adds numbers. Every option it sees already sums exactly, to the cent.

No call sees the whole month. Bank lines that pick the same entries become one group.

3 · BENCHMARK DESIGN

One fixed benchmark, exact gold, every model

60
months, 50 to 300 lines

10,000
bank lines

11,299
ledger entries

9
systems, 3 providers + local

Full runs

All 60 months: solver, GPT-4.1, GPT-OSS-120B, phi4 · GPT-5.1 on the 300-line months

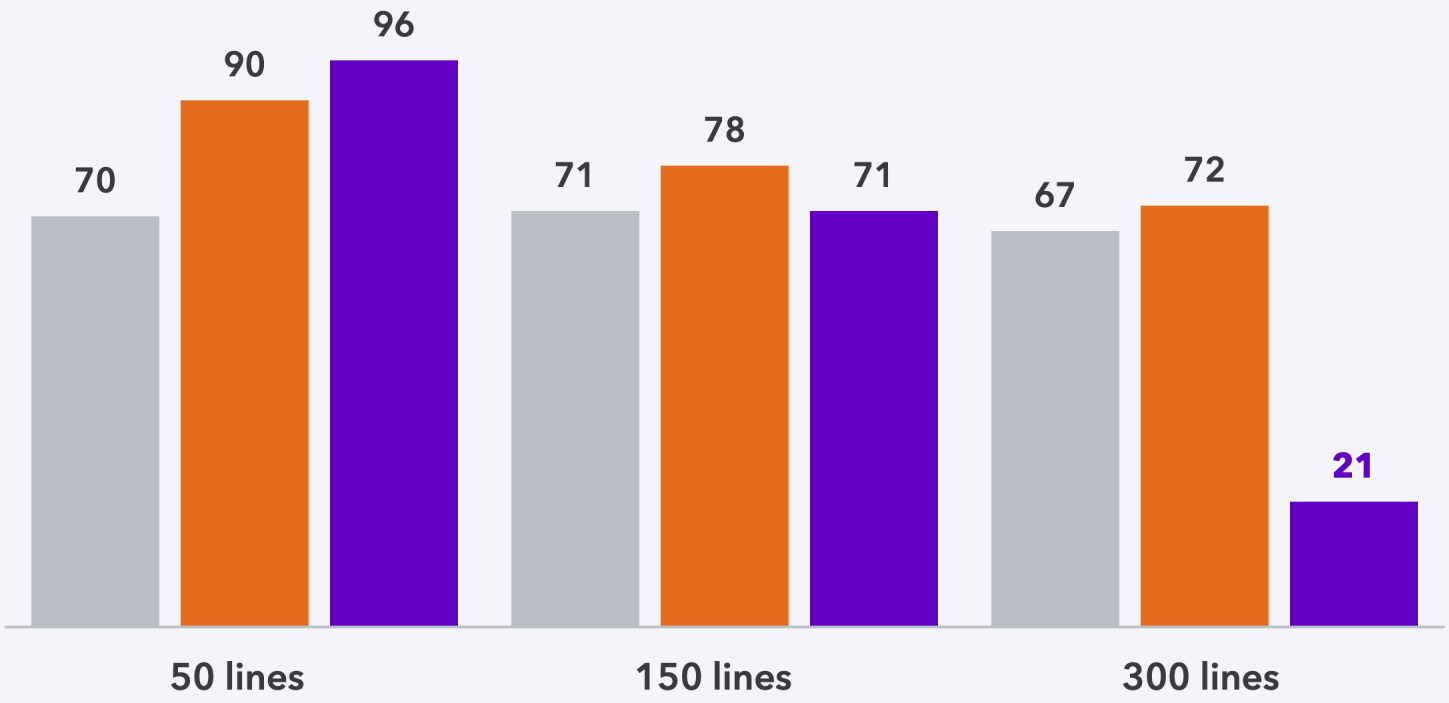
500-line cross-model set

One month per size, every model with and without the harness

Metric: exact group F1. A group counts only if every bank and ledger id matches gold. Plus precision, recall, line accuracy, recall by match type, false-accept rate, latency and cost.

3 · BASELINE

Whole-month prompting collapses as months grow



An amount-and-date solver beats GPT-OSS on whole months: 68.5% vs 49.3% over all 60.

Exact group F1 (%), all 60 hard months, 20 per size · GPT-OSS-120B at medium reasoning

3 · HEADLINE RESULT

49% → 96%

exact group F1, same model, same 60 months

21% → 94%
on 300-line months

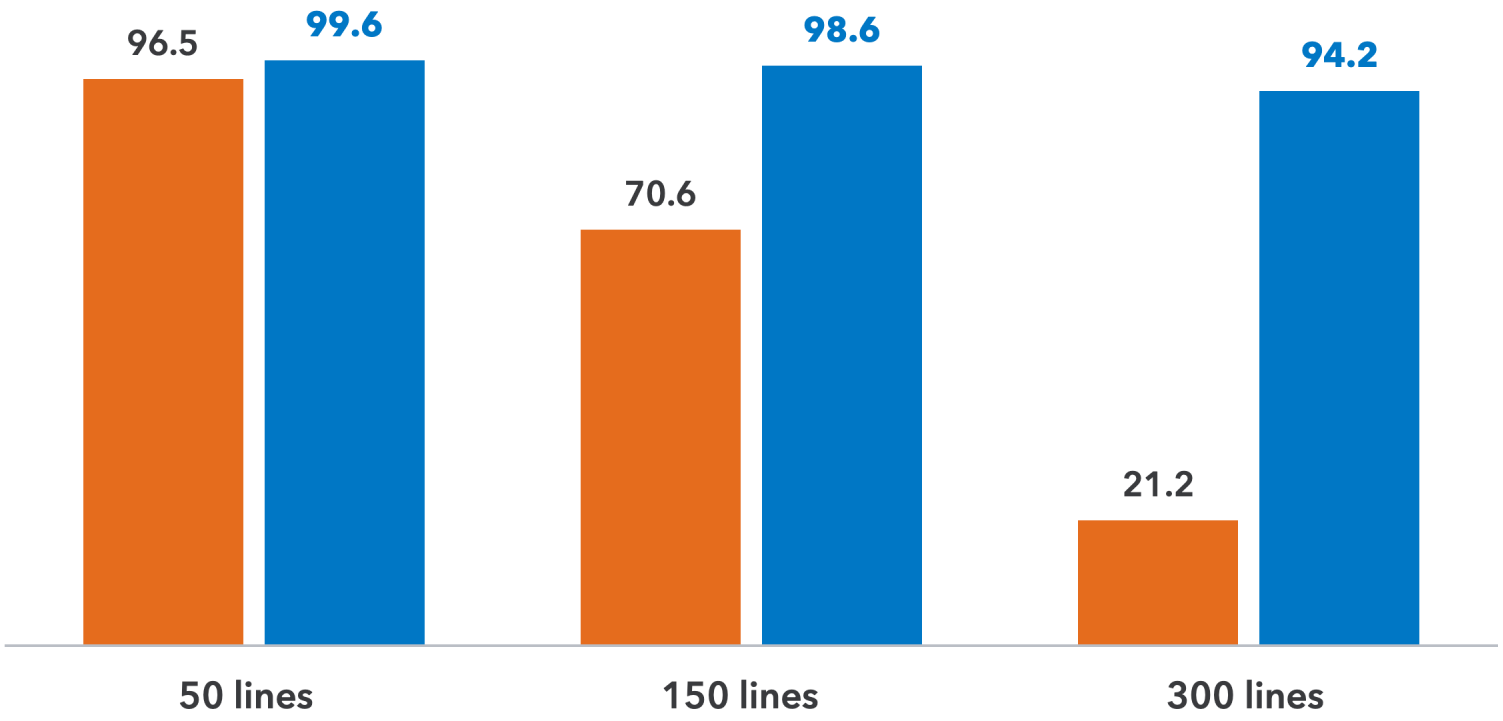
99.95%
precision on auto-accepted groups, 91.8% recall

$1.81
for all 10,000 lines

GPT-OSS-120B, medium reasoning, Fireworks serverless · whole month 49.31% → harness 96.06%

3 · RESULTS BY MONTH SIZE

The harness holds accuracy as months grow



Whole month

ReconIQ harness

+73 pts

at 300 lines, where whole-month prompting fails

Exact group F1 (%), GPT-OSS-120B medium reasoning, all 60 months (20 per size)

3 · ACROSS MODELS

Every model gains; smaller models gain most

Model	Whole month	Harness	Lift
GPT-4.1	72.7%	97.9%	+25.2
GPT-OSS-120B (low)	53.6%	96.4%	+42.8
Phi-4 14B *	22.0%	93.7%	+71.7
Nemotron 30B-A3B †	18.8%	88.3%	+69.5
Llama 3.1 8B	4.6%	61.5%	+56.9
Solver, no model	71.0%	–	–

8×

cheaper: open GPT-OSS within 1.5 pts of GPT-4.1, $0.12 vs $0.99 per 1,000 lines

Exact group F1, 500-line set (m50_00, m150_00, m300_00) · * 50 and 150 lines: 300 exceeds its 16k context · † harness mixes reasoning on and off

Four techniques, each fixing a failure mode

Technique	What it does	Recall: whole → harness
1 · Exact-sum enumeration	All ledger subsets of ≤4 that sum to the cent	1:many 2% → 100% · fee 6% → 100%
2 · Bounded choice	≤10 aliased candidates, pre-computed options	Decoys 48% → 100%
3 · Partial targets + assembly	Lines picking one entry merge into a group	Many:1 0% → 36%
4 · Sum, date, uniqueness checks	Auto-accept only groups that pass all three	99.95% precision accepted

Repair turns for baselines: checker feedback, empty answers flagged, context cuts detected

Reasoning control: 8 settings tested per model; runaway reasoning switched off

Group recall by match type, GPT-OSS-120B low reasoning, 500-line set · precision: medium reasoning, all 60 months

3 · EVALUATION RIGOR

Tested against the strongest baseline we had

+38.7 to +54.1

95% paired bootstrap CI on the F1 gain (points), by month · sign test $p = 8 \times 10^{-16}$

94.2% vs 97.2%

Open GPT-OSS in the harness vs GPT-5.1 reasoning on whole 300-line months, at ~6× lower cost

1 scorer

every run re-scored from its raw per-line log, the same way

5 × 2

match types × precision and recall, reported for every run, failures included

LIMITS AND NEXT STEPS

Where the remaining accuracy is

95.2%

Shortlist ceiling at 300 lines

Wider fee windows, larger cap on ambiguous lines

Many:1

Weakest group type

Rank partial targets higher; let the solver propose them

Real data

Synthetic benchmark so far

Frozen real-bank test set and a review workflow

The takeaway: constrain the model with code, verify its output, and a cheap open model reconciles at frontier accuracy.